

DG Interns Hub – Cybersecurity Intern- ship

Week 1: Assignment: Networking Fundamentals

Presented By : Prakhar gupta

Date: 6/9/2026

OSI Model

The **OSI (Open Systems Interconnection) Model** explains how data travels between devices over a network. It consists of **7 layers**, with each layer performing a specific function.

1. Physical Layer : Transmits raw bits through cables and signals.

Example: Ethernet cable

2. Data Link Layer : Provides node-to-node data transfer.

Example: MAC address, Switch

3. Network Layer : Handles routing and IP addressing.

Example: Router, IP address

4. Transport Layer : Provides end-to-end and reliable data delivery.

Example: TCP, UDP

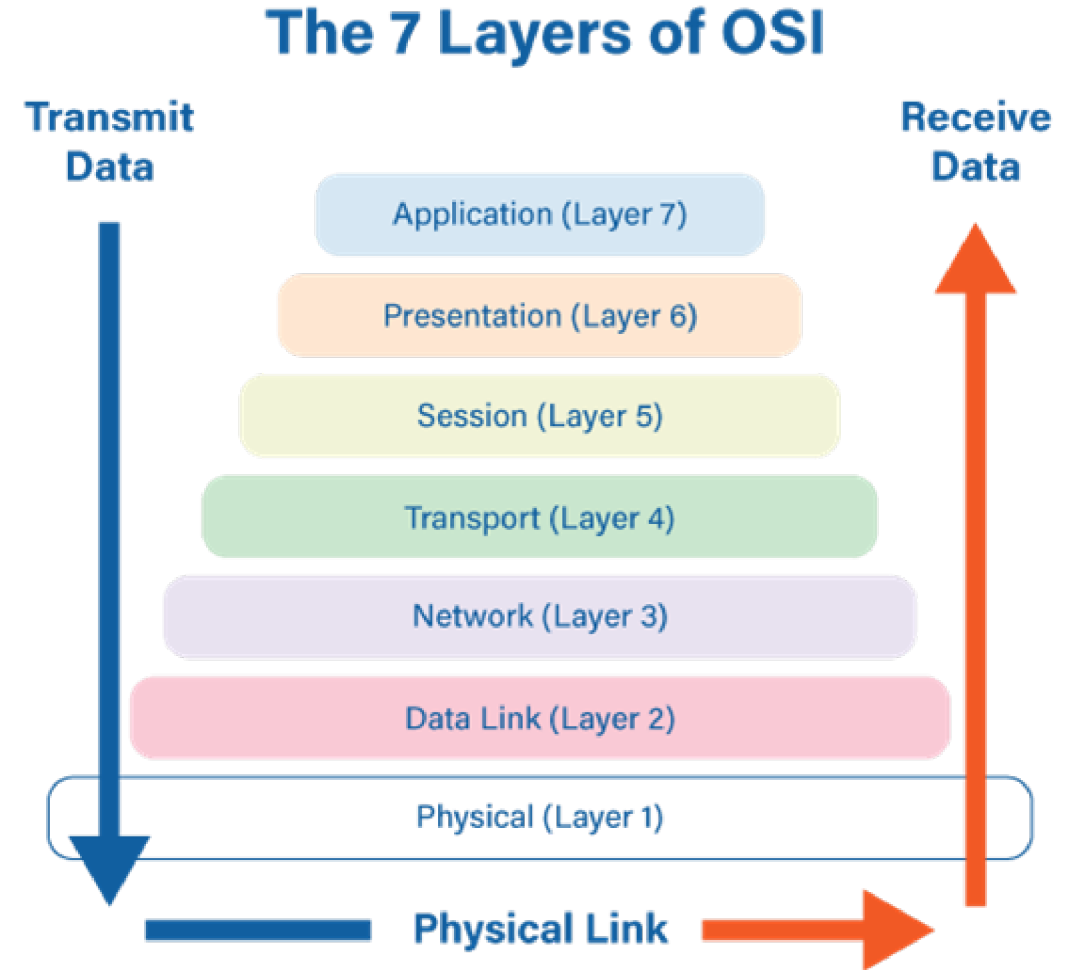
5. Session Layer : Establishes, manages, and terminates communication sessions.

Example: Login session

6. Presentation Layer : Handles data formatting, encryption, and compression.

Example: Data encryption

7. Application Layer : Provides network services to applications.**Example:** Web browser, HTTP/HTTPS



TCP/ IP Model

The **TCP/IP (Transmission Control Protocol/Internet Protocol) Model** is a practical networking model used for communication over the Internet. It consists of **4 layers**.

1. Application Layer

- Provides network services to applications.
- Handles communication between applications and network services.
- **Example:** HTTP, HTTPS, DNS, FTP

2. Transport Layer

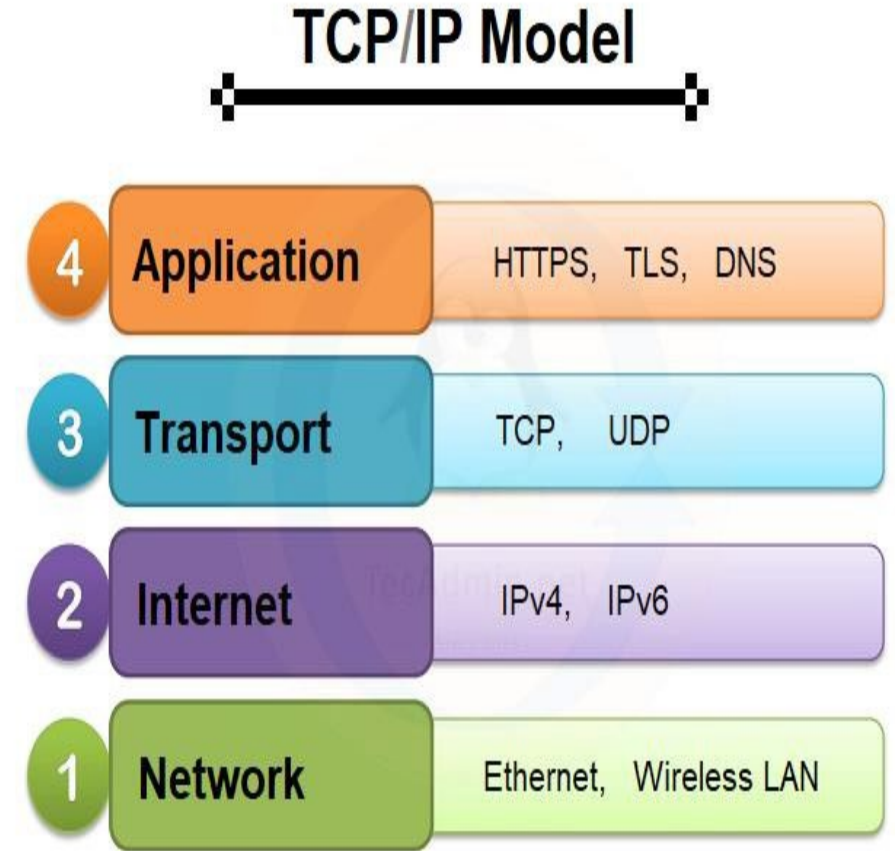
- Provides end-to-end communication between devices.
- Controls how data is delivered between applications.
- **Example:** TCP, UDP

3. Internet Layer

- Handles IP addressing and routing of packets.
- Determines how data travels between different networks.
- **Example:** IP, Router

4. Network Access Layer

- Handles physical transmission and local network communication.
- Transfers data between devices on the same network.



NETWORK PROTOCOLS

Network protocols are a set of rules that allow devices and applications to communicate and exchange data over a network.

1. HTTP / HTTPS

- Used for communication between web browsers and web servers.
- **HTTPS** provides encrypted and secure communication.
- **Example:** Opening a website in a web browser.

2. FTP

- Used to transfer files between computers over a network.
- **Example:** Uploading or downloading files from a server.

3. DNS

- Converts domain names into IP addresses.
- **Example:** google.com → IP address

4. DHCP

- Automatically assigns IP addresses and other network settings to devices.
- **Example:** Connecting a laptop to Wi-Fi.

IP Addressing

An **IP Address** is a unique numerical address assigned to a device on a network. It helps identify devices and enables communication between them.

Example

192.168.1.10

Public IP

- Used to identify a device or network on the Internet.
- Usually assigned by an Internet Service Provider (ISP).
- Can be used for communication over the Internet.

Private IP

- Used inside local networks.
- Common private IP ranges include: ○ **10.0.0.0 – 10.255.255.255** ○ **172.16.0.0 – 172.31.255.255** ○ **192.168.0.0 – 192.168.255.255**

Basic Subnetting

Subnetting divides a network into **smaller networks**.

Example:

- **IP Address:** 192.168.1.10
- **Subnet Mask:** 255.255.255.0

Network Commands

Practical: Network Commands

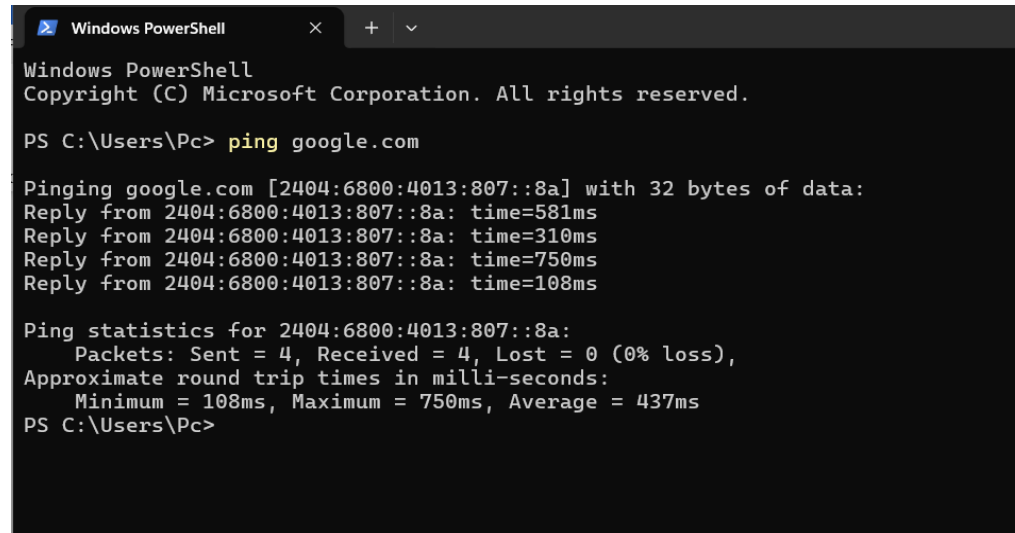
I used basic networking commands in Windows CMD to understand and troubleshoot network connectivity.

1.Ping ping

google.com

Checks connectivity between my computer and the destination.

➤ ping google.com

A screenshot of a Windows PowerShell terminal window. The title bar shows 'Windows PowerShell' with standard window controls. The terminal text is as follows:

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Pc> ping google.com

Pinging google.com [2404:6800:4013:807::8a] with 32 bytes of data:
Reply from 2404:6800:4013:807::8a: time=581ms
Reply from 2404:6800:4013:807::8a: time=310ms
Reply from 2404:6800:4013:807::8a: time=750ms
Reply from 2404:6800:4013:807::8a: time=108ms

Ping statistics for 2404:6800:4013:807::8a:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 108ms, Maximum = 750ms, Average = 437ms
PS C:\Users\Pc>
```

2.IP Configuration ipconfig

Displays the device's IP address, subnet mask and default gateway

➤ ipconfig

```
Windows PowerShell
PS C:\Users\Pc> ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 2:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::a8ff:50df:9519:f228%16
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : 
    IPv6 Address. . . . . : 2409:40e3:5429:9d21:aa6e:242e:fde0:88c3
    Temporary IPv6 Address. . . . . : 2409:40e3:5429:9d21:74d7:1771:4c48:657a
    Link-local IPv6 Address . . . . . : fe80::595f:72c9:ec7a:4822%15
    IPv4 Address. . . . . : 172.26.43.197
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::6024:4aff:fe89:b6b8%15
                                172.26.43.54

Ethernet adapter Bluetooth Network Connection:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
PS C:\Users\Pc> |
```

3.Traceroute

tracert google.com

Shows the path

takenby packets

from my computer
to the destination.

➤ `tracert google.com`

Cisco Packet Tracer

Aim:

```
Pinging google.com [2404:6800:4013:807::8a] with 32 bytes of data:
Reply from 2404:6800:4013:807::8a: time=581ms
Reply from 2404:6800:4013:807::8a: time=310ms
Reply from 2404:6800:4013:807::8a: time=750ms
Reply from 2404:6800:4013:807::8a: time=108ms

Ping statistics for 2404:6800:4013:807::8a:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 108ms, Maximum = 750ms, Average = 437ms
PS C:\Users\Pc> tracert google.com

Tracing route to google.com [2404:6800:4013:800::64]
over a maximum of 30 hops:

  1         2 ms         4 ms         2 ms      2409:40e3:5429:9d21::26
  2         *           *           *          Request timed out.
  3        419 ms       960 ms       417 ms      2405:200:5205:2f:d925::ff21
  4        147 ms       172 ms       313 ms      2405:200:80a:1513:61::6
  5         *           *           *          Request timed out.
  6        241 ms       319 ms        89 ms      2405:200:801:200::9a04
  7         *           *           *          Request timed out.
  8         *           *           *          Request timed out.
  9        424 ms       651 ms       269 ms      2001:4860:1:1::f48
 10        728 ms        *          681 ms      2404:6800:81e2:380::1
 11        238 ms        62 ms        47 ms      2001:4860:0:1::53ce
 12        328 ms        43 ms        50 ms      2001:4860:0:1::77dc
 13        235 ms        64 ms        41 ms      2001:4860:0:1::7841
 14        288 ms       411 ms       313 ms      2001:4860:0:1::8e0a
 15         *           *           *          Request timed out.
 16         *           *           *          Request timed out.
 17         *           *           *          Request timed out.
 18         *           *           *          Request timed out.
 19         *           *           *          Request timed out.
 20         *           *           *          Request timed out.
 21       443 ms       151 ms        56 ms      nu-in-f100.1e100.net [2404:6800:4013:800::64]

Trace complete.
PS C:\Users\Pc> |
```

To create and test a simple computer network using Cisco
Packet Tracer.

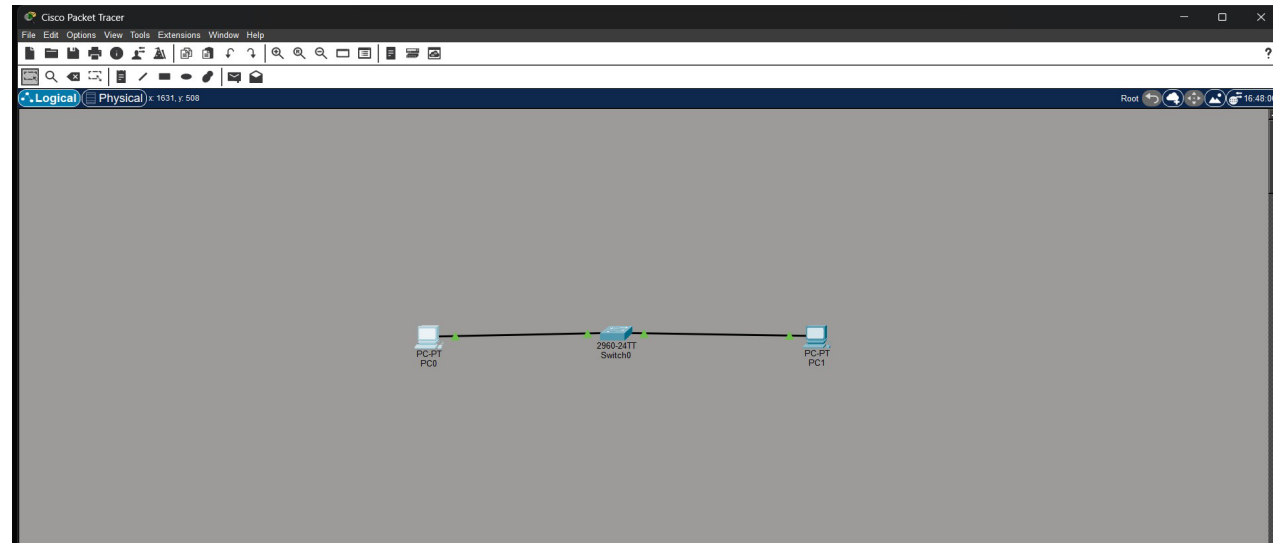
Network Components:

Device	IP Address	Subnet Mask
PC0	192.168.1.10	255.255.255.0
PC1	192.168.1.20	255.255.255.0

-  PC0 •  PC1
-  1 Switch
-  Ethernet cables
-  **Procedure**
- Open **Cisco Packet Tracer**.
- Add two PCs and one switch.
- Connect both PCs to the switch using **Copper Straight-Through** cables.
- Configure the IP address of each PC.
- Open **Command Prompt** on PC0.
- Use the command: ping 192.168.1.20
- Check whether PC0 receives a reply from PC1.

Result: PC0 successfully communicates with PC1 through the switch

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.20
Pinging 192.168.1.20 with 32 bytes of data:
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Reply from 192.168.1.20: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.1.20:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
```



Learnings & Conclusion

WEEK 1 Learning :

During the Networking Assignment, I learned:

- Fundamentals of computer networking.
- The **7 layers of the OSI model**.
- The **4 layers of the TCP/IP model**.
- Important protocols such as **HTTP, HTTPS, FTP, DNS and DHCP**.
- Basics of **IP addressing and subnetting**.
- Practical network commands such as **ping, ipconfig and tracert**.
- How to create and configure a network using **Cisco Packet Tracer**.
- How to test **network connectivity** using ping.

➤ **Conclusion**

This assignment strengthened my understanding of **networking, IP addressing, protocols, and troubleshooting**, providing a solid foundation for **cybersecurity and network security**.